

Study of Projectile Motion

Course: Engineering Dynamics

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Course Coordinator: Professor Firoz Alam

Name: Hugh Williams

Student Number:



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ABSTRACT

This lab compared the theoretical predictions made for projectile motion to the real-world experimental results. In this lab the time of flight, initial velocity and travelled horizontal distance.

INTRODUCTION

Projectile motion is the 2D movement of an object as it becomes airborne and its movements under the effect of gravity and its initial forces. This experiment was completed to investigate the principles of 2D kinematics through analysing features of projectile motion on a steel ball being accelerated from a PASCO mini launcher at different angles and initial velocities. The objectives of this investigation were to observe how conditions of launch effect airtime, how the speed and launch angle effects the horizontal travel distance, as well as how these can be theoretically predicted and the limitations that causes differences in the theoretical and experimental results. Horizontal displacement (x) can be theoretically determined through the equation, $x = v_{0x}t$, as the assumption of no horizontal forces such as air resistance means that the ball has no acceleration along the x-axis. However, in the vertical displacement (y) will accelerate due to gravity, while also assuming no other external forces such as air resistance are acting upon the ball, giving us its equation, $y = v_{0y}t - \frac{1}{2}gt^2$. We can determine the initial x and y launch velocities through the two equations, $v_{0x} = v_0 \cos(\theta)$ and $v_{0y} = v_0 \sin(\theta)$, where v_0 is the initial launch velocity and θ is the launch angle. The acceleration of gravity is additionally assumed to have a magnitude of 9.81 m/s^2 . The horizontal distance the equation used is $R = v_{0x} \times t$, where R is the theoretical horizontal displacement of the launched ball.

MATERIALS & METHOD

(Alam, Allhibi, & Hamed Allhibi 2021)

Materials

- Pasco Mini Launcher
- Photogate sensors
- Time-of-flight accessory
- Power Supply
- Universal interface
- Cables
- Retort stand
- C-Clamp
- Metal Ball
- Blunt-headed rod
- Carbon Paper
- Piece of blank white paper
- Measuring tape
- Computer
- Pasco capstone software
- Safety goggles

Method

Setup

1. Dress in PPE before starting any setup.
2. Secure the retort stand to bench with c-clamp.
3. Mount PASCO mini launcher on the retort stand.
4. Connect the closest photogate to the launcher into digital input 1 and the furthest to digital input 2.
5. Connect the time-of-flight accessory to digital input 3.
6. Connect the power supply and the universal interface.
7. Turn on by holding the power button until light comes on.
8. On the computer run the Pasco Capstone software.
9. Configure connections in the hardware setup panel as per the accessories input connections.
10. Select the timer setup and configure it to that shown in appendix 1.
11. Select measurements for time of flight and initial speed and assign them two-digit displays.

Procedure

1. Set launch angle of 0° , 30° or 60° with protractor on PASCO mini launcher.
2. Set launch range of 1 click (short), 2 clicks (medium) or 3 clicks (long) and accordingly adjust the move the time-of-flight accessory to sit where the ball will land.
3. Place ball in launcher with assistance of the blunt-head rod.
4. Start recording on the Pasco Capstone Software before the ball is launch.
5. Pull trigger to launch ball.
6. Note down the velocity and flight time recorded by the software.
7. Using a measuring tape, record the horizontal displacement.
8. Repeat from step 1 for each range and each launch angle.

RESULTS

Raw Data

0° (Horizontal) Launch	Launch Height (cm)	Pad Height (cm)	Range	Trial	Distance (cm)	Time (s)	Initial Velocity (m/s)
	48	2.5	Short (1 Click)	1	87.5	0.29	3.04
	48	2.5	Short (1 Click)	2	87.5	0.29	3.04
	48	2.5	Short (1 Click)	3	87.5	0.29	3.02
	48	2.5	Medium (2 Clicks)	1	117	0.29	4.01
	48	2.5	Medium (2 Clicks)	2	117	0.29	4.03
	48	2.5	Medium (2 Clicks)	3	117	0.29	4.01
	48	2.5	Long (3 Clicks)	1	154	0.29	5.32
	48	2.5	Long (3 Clicks)	2	154	0.29	5.43
	48	2.5	Long (3 Clicks)	3	154	0.29	5.30
30° Launch	Launch Height	Pad Height	Range	Trial	Distance (cm)	Time (s)	Initial Velocity (m/s)
	48	2.5	Short (1 Click)	1	129.2	0.50	2.91
	48	2.5	Short (1 Click)	2	129.2	0.50	2.93
	48	2.5	Short (1 Click)	3	129.2	0.50	2.93
60° Launch	Launch Height	Pad Height	Range	Trial	Distance (cm)	Time (s)	Initial Velocity (m/s)
	50	2.5	Short (1 Click)	1	100	0.68	2.87
	50	2.5	Short (1 Click)	2	100	0.68	2.97
	50	2.5	Short (1 Click)	3	100	0.68	2.87

Table 1, Experiment Data Table

Calculations

Short Range (1 Click) - 0° Launch

$$v_{0x} = v_0, \quad v_{0y} = 0$$

$$v_{0x} = 3.04 \text{ m/s}$$

$$R_{theoretical} = v_{0x} \times t = 3.04 \times 0.29 = 0.8816 \text{ m}$$

$$R_{theoretical} = 88.16 \text{ cm}$$

$$R_{measured} = 87.5 \text{ cm}$$

$$\%error = \frac{|87.5 - 88.16|}{88.16} \times 100 = 0.75\%$$

$$H_{max} = H_0 = 0.48 \text{ m} = 48 \text{ cm}$$

$$y = H_0 + v_{0y}t - \frac{1}{2}gt^2$$

$$H_f = \text{landing height} = 0.025 \text{ m} = 2.5 \text{ cm}$$

$$t_{measured} = 0.29 \text{ s}$$

$$t = 0.00, \quad y = 0.48 + 0(0.00) - \frac{1}{2} \times 9.81 \times (0.00)^2 = 0.48 \text{ m} = 48 \text{ cm}$$

$$t = 0.06, \quad y = 0.48 + 0(0.06) - \frac{1}{2} \times 9.81 \times (0.06)^2 = 0.4623 \text{ m} = 46.23 \text{ cm}$$

$$t = 0.12, \quad y = 0.48 + 0(0.12) - \frac{1}{2} \times 9.81 \times (0.12)^2 = 0.4094 \text{ m} = 40.94 \text{ cm}$$

$$t = 0.18, \quad y = 0.48 + 0(0.18) - \frac{1}{2} \times 9.81 \times (0.18)^2 = 0.3211 \text{ m} = 32.11 \text{ cm}$$

$$t = 0.24, \quad y = 0.48 + 0(0.24) - \frac{1}{2} \times 9.81 \times (0.24)^2 = 0.1975 \text{ m} = 19.75 \text{ cm}$$

$$t = 0.29, \quad y = 0.48 + 0(0.29) - \frac{1}{2} \times 9.81 \times (0.29)^2 = 0.0675 \text{ m} = 6.75 \text{ cm}$$

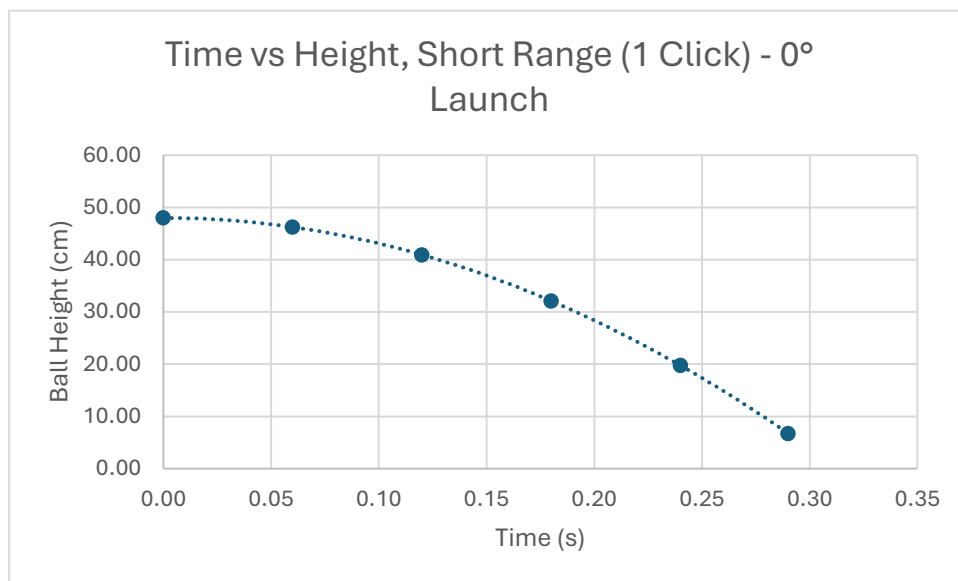
$$H_f < y, \text{ when } t = 0.29$$

$$\therefore t_{\text{theoretical}} \neq 0.29 \text{ s}$$

$$\text{Let } y = H_f \Rightarrow 0.025 = 0.48 + 0(t) - \frac{1}{2} \times 9.81 \times t^2$$

$$t_{\text{theoretical}} = 0.305 \text{ s}$$

$$\%_{t/\text{error}} = \frac{|t_{\text{theoretical}} - t_{\text{measured}}|}{t_{\text{theoretical}}} \times 100 = \frac{|0.305 - 0.29|}{0.305} \times 100 = 4.92\%$$



Graph 1

Theoretical Distance (cm)	Maximum Height (cm)	Initial Horizontal Velocity (m/s)	Initial Vertical Velocity (m/s)	Distance Error (%)	Theoretical Time (s)	Time Error (%)
88.16	48	3.04	0	0.75	0.305	4.92

Medium Range (2 Clicks) - 0° Launch

$$v_{0x} = v_0, \quad v_{0y} = 0$$

$$v_{0x} = 4.01 \text{ m/s}$$

$$R_{\text{theoretical}} = v_{0x} \times t = 4.01 \times 0.29 = 1.1629 \text{ m}$$

$$R_{\text{theoretical}} = 116.29 \text{ cm}$$

$$R_{\text{measured}} = 117 \text{ cm}$$

$$\%_{\text{error}} = \frac{|117 - 116.29|}{116.29} \times 100 = 0.61\%$$

$$H_{\text{max}} = H_0 = 0.48 \text{ m} = 48 \text{ cm}$$

$$y = H_0 + v_{0y}t - \frac{1}{2}gt^2$$

$$H_f = \text{landing height} = 0.025 \text{ m} = 2.5 \text{ cm}$$

$$t_{\text{measured}} = 0.29 \text{ s}$$

$$t = 0.00, \quad y = 0.48 + 0(0.00) - \frac{1}{2} \times 9.81 \times (0.00)^2 = 0.48 \text{ m} = 48 \text{ cm}$$

$$t = 0.06, \quad y = 0.48 + 0(0.06) - \frac{1}{2} \times 9.81 \times (0.06)^2 = 0.4623 \text{ m} = 46.23 \text{ cm}$$

$$t = 0.12, \quad y = 0.48 + 0(0.12) - \frac{1}{2} \times 9.81 \times (0.12)^2 = 0.4094 \text{ m} = 40.94 \text{ cm}$$

$$t = 0.18, \quad y = 0.48 + 0(0.18) - \frac{1}{2} \times 9.81 \times (0.18)^2 = 0.3211 \text{ m} = 32.11 \text{ cm}$$

$$t = 0.24, \quad y = 0.48 + 0(0.24) - \frac{1}{2} \times 9.81 \times (0.24)^2 = 0.1975 \text{ m} = 19.75 \text{ cm}$$

$$t = 0.29, \quad y = 0.48 + 0(0.29) - \frac{1}{2} \times 9.81 \times (0.29)^2 = 0.0675 \text{ m} = 6.75 \text{ cm}$$

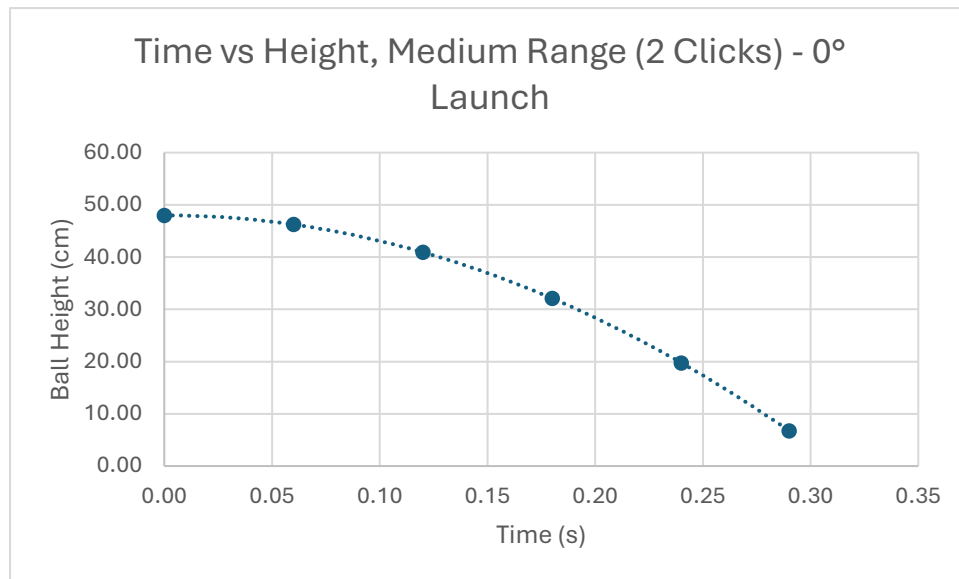
$$H_f < y, \text{ when } t = 0.29$$

$$\therefore t_{\text{theoretical}} \neq 0.29 \text{ s}$$

$$\text{Let } y = H_f \Rightarrow 0.025 = 0.48 + 0(t) - \frac{1}{2} \times 9.81 \times t^2$$

$$t_{\text{theoretical}} = 0.305 \text{ s}$$

$$\%_{t/\text{error}} = \frac{|t_{\text{theoretical}} - t_{\text{measured}}|}{t_{\text{theoretical}}} \times 100 = \frac{|0.305 - 0.29|}{0.305} \times 100 = 4.92\%$$



Graph 2

Theoretical Distance (cm)	Maximum Height (cm)	Initial Horizontal Velocity (m/s)	Initial Vertical Velocity (m/s)	Distance Error (%)	Theoretical Time (s)	Time Error (%)
116.29	48	4.01	0	0.61	0.305	4.92

Long Range (3 Clicks) - 0° Launch

$$v_{0x} = v_0, \quad v_{0y} = 0$$

$$v_{0x} = 5.32 \text{ m/s}$$

$$R_{\text{theoretical}} = v_{0x} \times t = 5.32 \times 0.29 = 1.5428 \text{ m}$$

$$R_{\text{theoretical}} = 154.28 \text{ cm}$$

$$R_{\text{measured}} = 154 \text{ cm}$$

$$\%_{\text{error}} = \frac{|154 - 154.28|}{154.28} \times 100 = 0.18\%$$

$$H_{\text{max}} = H_0 = 0.48 \text{ m} = 48 \text{ cm}$$

$$y = H_0 + v_{0y}t - \frac{1}{2}gt^2$$

$$H_f = \text{landing height} = 0.025 \text{ m} = 2.5 \text{ cm}$$

$$t_{\text{measured}} = 0.29 \text{ s}$$

$$t = 0.00, \quad y = 0.48 + 0(0.00) - \frac{1}{2} \times 9.81 \times (0.00)^2 = 0.48 \text{ m} = 48 \text{ cm}$$

$$t = 0.06, \quad y = 0.48 + 0(0.06) - \frac{1}{2} \times 9.81 \times (0.06)^2 = 0.4623 \text{ m} = 46.23 \text{ cm}$$

$$t = 0.12, \quad y = 0.48 + 0(0.12) - \frac{1}{2} \times 9.81 \times (0.12)^2 = 0.4094 \text{ m} = 40.94 \text{ cm}$$

$$t = 0.18, \quad y = 0.48 + 0(0.18) - \frac{1}{2} \times 9.81 \times (0.18)^2 = 0.3211 \text{ m} = 32.11 \text{ cm}$$

$$t = 0.24, \quad y = 0.48 + 0(0.24) - \frac{1}{2} \times 9.81 \times (0.24)^2 = 0.1975 \text{ m} = 19.75 \text{ cm}$$

$$t = 0.29, \quad y = 0.48 + 0(0.29) - \frac{1}{2} \times 9.81 \times (0.29)^2 = 0.0675 \text{ m} = 6.75 \text{ cm}$$

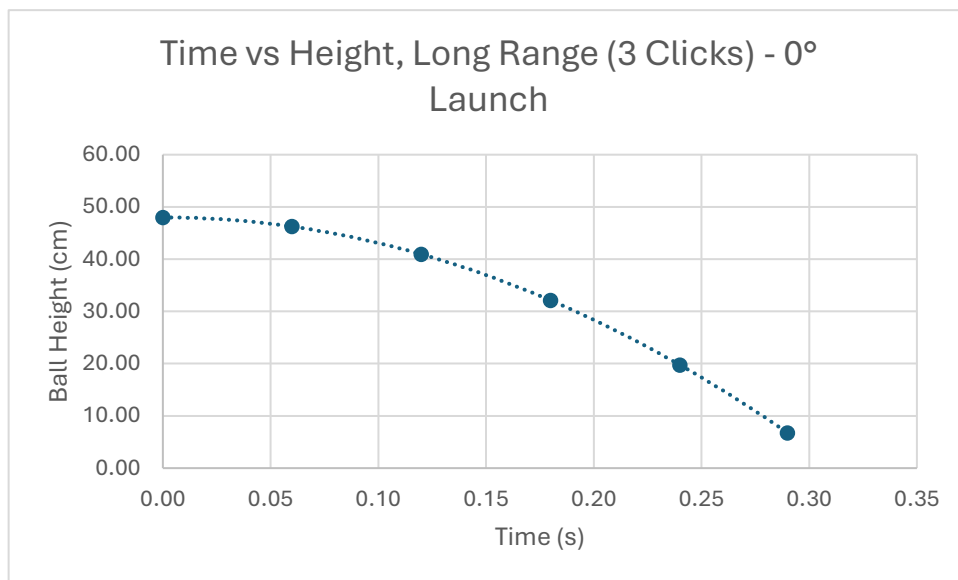
$$H_f < y, \text{ when } t = 0.29$$

$$\therefore t_{\text{theoretical}} \neq 0.29 \text{ s}$$

$$\text{Let } y = H_f \Rightarrow 0.025 = 0.48 + 0(t) - \frac{1}{2} \times 9.81 \times t^2$$

$$t_{\text{theoretical}} = 0.305 \text{ s}$$

$$\%_{t/\text{error}} = \frac{|t_{\text{theoretical}} - t_{\text{measured}}|}{t_{\text{theoretical}}} \times 100 = \frac{|0.305 - 0.29|}{0.305} \times 100 = 4.92\%$$



Graph 3

Theoretical Distance (cm)	Maximum Height (cm)	Initial Horizontal Velocity (m/s)	Initial Vertical Velocity (m/s)	Distance Error (%)	Theoretical Time (s)	Time Error (%)
154.28	48	5.32	0	0.18	0.305	4.92

Short Range (1 Click) - 30° Launch

$$v_{0x} = v_0 \cos \theta = 2.91 \times \cos 30 = 2.520 \text{ m/s}$$

$$v_{0y} = v_0 \sin \theta = 2.91 \times \sin 30 = 1.455 \text{ m/s}$$

$$R_{theoretical} = v_{0x} \times t = 2.520 \times 0.50 = 1.260 \text{ m}$$

$$R_{theoretical} = 126.0 \text{ cm}$$

$$R_{measured} = 129.2 \text{ cm}$$

$$\%_{error} = \frac{|129.2 - 126.0|}{126.0} \times 100 = 2.54\%$$

$$H_{max} = H_0 + H_{rise} = H_0 + \frac{v_{0y}^2}{2g} = 0.48 + \frac{1.455^2}{2 \times 9.81} = 0.5879 \text{ m}$$

$$H_{max} = 58.79 \text{ cm}$$

$$H_0 = 0.48 \text{ m} = 48 \text{ cm}$$

$$y = H_0 + v_{0y}t - \frac{1}{2}gt^2$$

$$H_f = \text{landing height} = 0.025 \text{ m} = 2.5 \text{ cm}$$

$$t_{measured} = 0.29s$$

$$t = 0.00, \quad y = 0.48 + 1.455 \times (0.00) - \frac{1}{2} \times 9.81 \times (0.00)^2 = 0.48 \text{ m} = 48 \text{ cm}$$

$$t = 0.10, \quad y = 0.48 + 1.455 \times (0.10) - \frac{1}{2} \times 9.81 \times (0.10)^2 = 0.5765 \text{ m} = 57.65 \text{ cm}$$

$$t = 0.20, \quad y = 0.48 + 1.455 \times (0.20) - \frac{1}{2} \times 9.81 \times (0.20)^2 = 0.5748 \text{ m} = 57.48 \text{ cm}$$

$$t = 0.30, \quad y = 0.48 + 1.455 \times (0.30) - \frac{1}{2} \times 9.81 \times (0.30)^2 = 0.4751 \text{ m} = 47.51 \text{ cm}$$

$$t = 0.40, \quad y = 0.48 + 1.455 \times (0.40) - \frac{1}{2} \times 9.81 \times (0.40)^2 = 0.2772 \text{ m} = 27.72 \text{ cm}$$

$$t = 0.50, \quad y = 0.48 + 1.455 \times (0.50) - \frac{1}{2} \times 9.81 \times (0.50)^2 = -0.0188 \text{ m} = -1.88 \text{ cm}$$

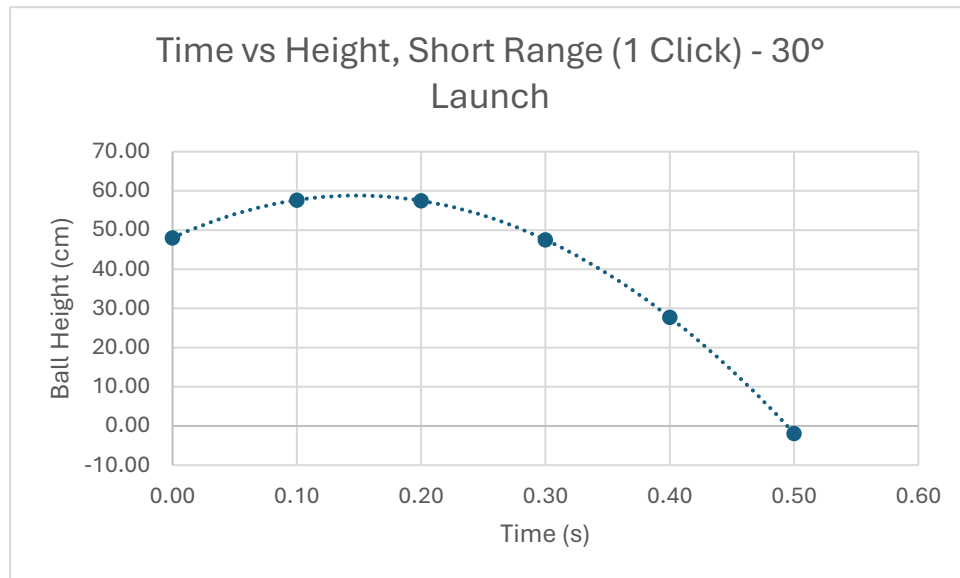
$$H_f > y, \text{ when } t = 0.50$$

$$\therefore t_{theoretical} \neq 0.50s$$

$$\text{Let } y = H_f \Rightarrow 0.025 = 0.48 + 1.455 \times (t) - \frac{1}{2} \times 9.81 \times t^2$$

$$t_{theoretical} = 0.495s$$

$$\%_{t/error} = \frac{|t_{theoretical} - t_{measured}|}{t_{theoretical}} \times 100 = \frac{|0.495 - 0.50|}{0.495} \times 100 = 1.01\%$$



Graph 4

Theoretical Distance (cm)	Maximum Height (cm)	Initial Horizontal Velocity (m/s)	Initial Vertical Velocity (m/s)	Distance Error (%)	Theoretical Time (s)	Time Error (%)
126	58.79	2.52	1.455	2.54	0.495	1.01

Short Range (1 Click) - 60° Launch

$$v_{0x} = v_0 \cos \theta = 2.87 \times \cos 60 = 1.435 \text{ m/s}$$

$$v_{0y} = v_0 \sin \theta = 2.87 \times \sin 60 = 2.485 \text{ m/s}$$

$$R_{theoretical} = v_{0x} \times t = 1.435 \times 0.68 = 0.9758 \text{ m}$$

$$R_{theoretical} = 97.58 \text{ cm}$$

$$R_{measured} = 100 \text{ cm}$$

$$\%_{error} = \frac{|100 - 97.58|}{97.58} \times 100 = 2.48\%$$

$$H_{max} = H_0 + H_{rise} = H_0 + \frac{v_{0y}^2}{2g} = 0.5 + \frac{2.485^2}{2 \times 9.81} = 0.8147 \text{ m}$$

$$H_{max} = 81.47 \text{ cm}$$

$$H_0 = 0.50 \text{ m} = 50 \text{ cm}$$

$$y = H_0 + v_{0y}t - \frac{1}{2}gt^2$$

$$H_f = \text{landing height} = 0.025 \text{ m} = 2.5 \text{ cm}$$

$$t_{measured} = 0.68 \text{ s}$$

$$t = 0.00, \quad y = 0.5 + 2.485 \times (0.00) - \frac{1}{2} \times 9.81 \times (0.00)^2 = 0.5 \text{ m} = 50 \text{ cm}$$

$$t = 0.10, \quad y = 0.5 + 2.485 \times (0.10) - \frac{1}{2} \times 9.81 \times (0.10)^2 = 0.6995 \text{ m} = 69.95 \text{ cm}$$

$$t = 0.20, \quad y = 0.5 + 2.485 \times (0.20) - \frac{1}{2} \times 9.81 \times (0.20)^2 = 0.8008 \text{ m} = 80.08 \text{ cm}$$

$$t = 0.30, \quad y = 0.5 + 2.485 \times (0.30) - \frac{1}{2} \times 9.81 \times (0.30)^2 = 0.8041 \text{ m} = 80.41 \text{ cm}$$

$$t = 0.40, \quad y = 0.5 + 2.485 \times (0.40) - \frac{1}{2} \times 9.81 \times (0.40)^2 = 0.7092 \text{ m} = 70.92 \text{ cm}$$

$$t = 0.50, \quad y = 0.5 + 2.485 \times (0.50) - \frac{1}{2} \times 9.81 \times (0.50)^2 = 0.5163 \text{ m} = 51.63 \text{ cm}$$

$$t = 0.60, \quad y = 0.5 + 2.485 \times (0.60) - \frac{1}{2} \times 9.81 \times (0.60)^2 = 0.2252 \text{ m} = 22.52 \text{ cm}$$

$$t = 0.68, \quad y = 0.5 + 2.485 \times (0.68) - \frac{1}{2} \times 9.81 \times (0.68)^2 = -0.0783 \text{ m} = -7.83 \text{ cm}$$

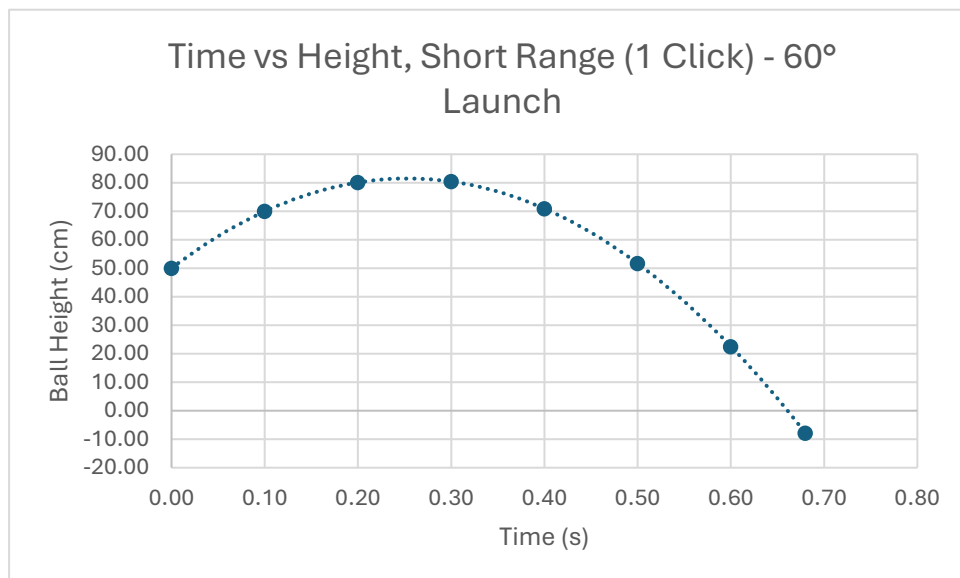
$$H_f > y, \text{ when } t = 0.68$$

$$\therefore t_{\text{theoretical}} \neq 0.68 \text{ s}$$

$$\text{Let } y = H_f \Rightarrow 0.025 = 0.48 + 2.485 \times (t) - \frac{1}{2} \times 9.81 \times t^2$$

$$t_{\text{theoretical}} = 0.661 \text{ s}$$

$$\%t_{\text{error}} = \frac{|t_{\text{theoretical}} - t_{\text{measured}}|}{t_{\text{theoretical}}} \times 100 = \frac{|0.661 - 0.68|}{0.661} \times 100 = 2.87\%$$



Graph 5

Theoretical Distance (cm)	Maximum Height (cm)	Initial Horizontal Velocity (m/s)	Initial Vertical Velocity (m/s)	Distance Error (%)	Theoretical Time (s)	Time Error (%)
97.58	81.47	1.435	2.485	2.48	0.661	2.87

DISCUSSION

The results portrayed a correlation between a greater launch angle and a greater flight time. A launch angle of 30° had greater time in the air than a 0° launch angle and when the launch angle was 60°, it had a greater airtime than the 30° launch angle. This aligns with the theory of projectile motion as with a greater launch angle, the vertical initial velocity (v_{0y}) is increased relative to the initial velocity. From the equation in *Appendix B* to finding the height of the ball at a point in time, we can see that a greater velocity increases the time it takes to decelerate to a zero vertical velocity (at the peak), as well as a higher time to reach the time-of-flight accessory. The horizontal launch (0°) had the smallest airtime as it didn't gain any height from launch and immediately started falling from its initial height. In the equation in *Appendix E*, we can see it is missing a section of the equation in *Appendix B*. Not having the initial vertical velocity outlines that the vertical position of the ball only has the vertical acceleration of gravity impacting its location.

From the results section we can see that the angle the ball is launched at effects the shape of its trajectory. At a 60° launch angle, relatively more of the initial velocity's force worked towards increasing the height of the ball than a 30° launch angle. This led to the trajectory being a steeper parabolic shape, while having a smaller horizontal displacement. Whilst the 30° launch angle doesn't have as great of a maximum height, the greater force from a relatively higher initial horizontal velocity results in a wider parabolic curve and a larger horizontal displacement. Additionally, we can support this through visualising it through quadratic equations. For this experiment it can be graphed with the equation in *Appendix C*. When plugging in the initial horizontal and vertical velocities as well as 9.81 for gravity, we get a graph for the trajectory of both launch angles, seen in *Appendix D*, reinforcing the relation between launch angle and change in trajectory.

The change in trajectory can also be viewed with a change the initial velocity. From the three trials performed at a 0°, it can be observed that the greater the initial velocity, the more elongated the parabolic curve. This can be graphed using the same equation in *Appendix C*. However, due to there being no initial vertical velocity, we can simplify it to the equation found in *Appendix E*. Graphing these for each initial velocity in *Appendix F*, again reinforces that while launch angle alters the trajectory, so does the initial velocity.

The measured horizontal range was relatively close to the theoretical range, with the smallest margin of error being 0.18% and the greatest margin of error being 2.54%. This suggests that the ball followed the expected kinematic behaviours, indicating that using projectile motion calculations allow for reasonable assumptions for reality. However, the discrepancies between the theoretical and measured results are unexpected. Initial beliefs were that the theoretical results would produce a greater distance than the measured results due to our assumptions ignoring air resistance and other external factors that would only impact real world tests. While this is unexpected, we can guess that human or machine errors are the most likely culprits for these discrepancies. Additionally, as the launch angle is increased from the horizontal tests, the

margin of error increases to around 2.5% from less than 1%. We can speculate from this shift that another factor in the differences between the theoretical and measured results is the increased sensitivity to uncertainties at greater trajectories.

Other sources of error include air resistance, launcher angle, machinery or tool error. Air resistance was assumed to be negligible for all calculation in this experiment. However, it was still present in the trials, impacting the measured results. Air resistance increases the drag on the ball, usually resulting in theoretical results to be slightly higher. The launcher angle is also a location where discrepancies may have occurred. In theoretical measurements it was assumed that the launcher was at an exact angle, small alignment errors are possible even with using a protractor, these would in turn alter both the vertical and horizontal initial velocities. Furthermore, mechanical errors such as with slight inconsistencies in the launcher or with delays in sensor triggering or signals causing deviations in the measured results.

This lab experiment did prove projectile motion theory with minimal deviations between measured and theoretical results. The relationship between time of flight, trajectory shape, launch angle and initial velocity can all be observed in both theoretical and lab conditions being not very dissimilar. While slight variations are there, they can easily be explained by the assumptions and limitations outlined above. Therefore, kinematic equations for predicting projectile motion can be deemed as reliable under similar circumstances and controlled conditions.

CONCLUSION

During this lab we were able to conclude that under the conditions of the lab projectile motion theory holds up to reality, proving it is appropriate to use in kinematic approximations. This was proved through the differing trial variables acting impacting outcomes in ways that the theory outlines. This includes the launch angle increasing the balls time airborne, and the range changing accurately with different launch speeds and launch angles. While there were differences between theoretical and measured results, they were minimal and can be attributed to assumptions or minimal errors.

SUGGESTIONS

Use a more accurate calibration method for the launcher angle. Whilst using the built-in protractor is relatively accurate, it is still subject to human errors. However, a launcher that can be changed with mechanical systems is more likely to offer the most accurate launch angle.

An increased trial count would also bolster support for projectile motion theory. In this experiment we only completed three trials per set, so the results we received from them have the chance to be outliers. Gathering larger dataset would allow for a more uniform data distribution which would allow for more reliable conclusions to be made.

Using an automated range measurement system would allow for the most accurate horizontal displacement. Using the current paper method is open to inaccuracies such as a smeared impact mark or unclear first impact point. Using an automated system would greatly minimise the human error involved in the measurements.

While the conditions for this test were mostly unaffected by external factors, they still occur such as with air resistance. Including an uncertainty analysis where these slight altering forces are accounted for would allow for a precise experiment that's produce more reliable results.

REFERENCES

Alam, F, Allhibi, A, & Hamed Allhibi 2021, *Experiment: Study of projectile motion*, RMIT University.

Catalyst, P 2020, "How to Calculate Percentage Error - Formula, Steps & Examples," *Physicscatalyst's Blog*, viewed <<https://physicscatalyst.com/article/percentage-error/>>.

APPENDICES

Appendix A

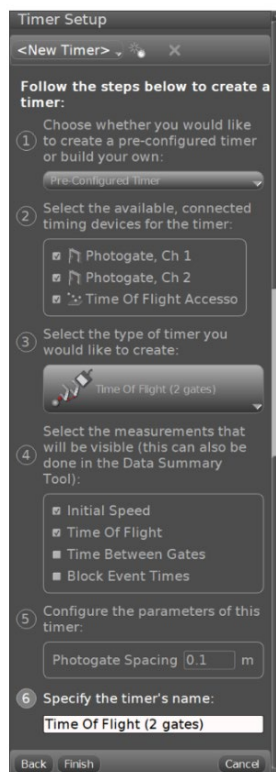


Figure 1, Cited from (Alam, Allhibi, & Hamed Allhibi 2021, sec. Appendix A)

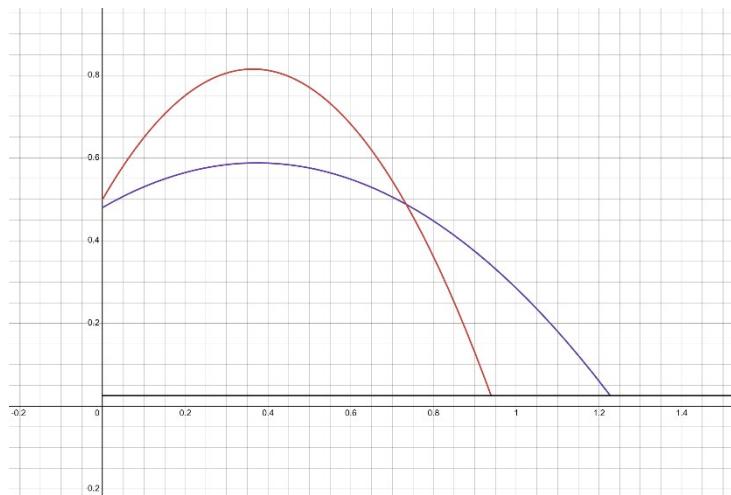
Appendix B

$$y = H_0 + v_{0y}t - \frac{1}{2}gt^2$$

Appendix C

$$y = H_0 + v_{0y}\left(\frac{x}{v_{0x}}\right) - \frac{1}{2}g\left(\frac{x}{v_{0x}}\right)^2$$

Appendix D

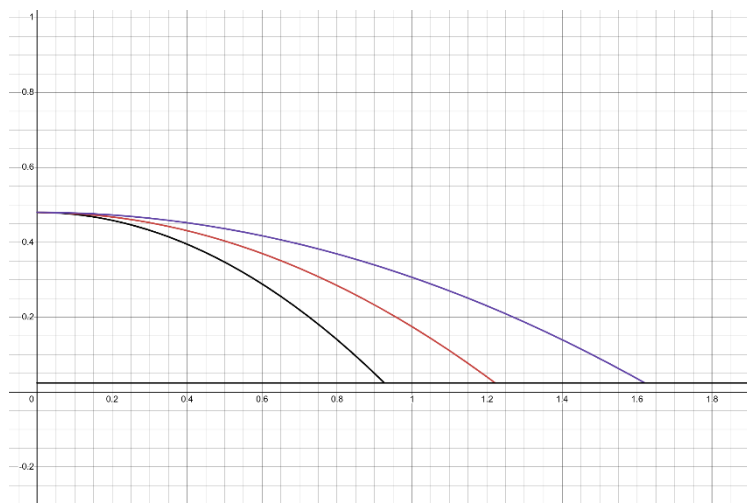


*Desmos Graph 1, Red Curve (Short Range (1 Click) - 60° Launch Angle)
| Purple Curve (Short Range (1 Click) - 30° Launch Angle)*

Appendix E

$$y = H_0 - \frac{1}{2}g\left(\frac{x}{v_{0x}}\right)^2$$

Appendix F



*Desmos Graph 2, All 0° launch Angle, Black Curve (Short Range (1 Click))
| Red Curve (Medium Range (2 Clicks)) | Purple Curve (Long Range (3 Clicks))*